

A Pose-Aware Global Representation Network for Human Parsing

Yanghong Zhou^{ID} and P. Y. Mok^{ID}, *Member, IEEE*

Abstract—Many recognition tasks including image/video classification, segmentation and object detection can be improved by the integration of global information. Although global information may be better represented in some recognition tasks than the others, it is worth exploring how global information from related tasks can be effectively used to improve the performance of a target task. The task of pose estimation predicts the locations of human joints, thus providing global information about the human body. In this paper, we propose a pose-aware global representation network model (PAGRnet) that exploits *global information* from pose estimation to enhance feature learning in human parsing. In our PAGRnet model, a novel learning module with three integrated parts is used to learn global information. The first part generates a global joint representation, while the second part learns the relationship between the pixels and joints. By integrating the global joint representation with the pixel-joint relationship, the resulting pose-aware global representation is augmented for the parsing task. Our experimental results show competitive performance of our method on the LIP, the Pascal-person-part and the ATR datasets, with reduced computation costs in comparison to other proposals of global information fusion. We also demonstrate the advantages of our feature fusion model over concatenation, pixel-wise and channel-wise relation models.

Index Terms—Global information fusion, human parsing, human pose estimation, pose-aware information fusion, semantic segmentation.

I. INTRODUCTION

THE aim of human parsing is to segment various body parts and clothing items from an input human image. Human parsing is an important task in computer vision, because it supports many human-centred applications, such as fashion analysis [1], [2], [3], person re-identification [4], [5], action recognition [6], [7], and pedestrian detection [8].

Manuscript received 25 May 2022; revised 9 August 2022 and 24 September 2022; accepted 2 October 2022. Date of publication 10 October 2022; date of current version 5 April 2023. This work was supported in part by the Research Grants Council of the Hong Kong Special Administrative Region, China, under Grant 152161/17E and Grant 152112/19E; in part by The Hong Kong Polytechnic University under Project ZVR3; and in part by the Laboratory for Artificial Intelligence in Design, Innovation and Technology Fund, Hong Kong Special Administrative Region, under Project RP1-1. This article was recommended by Associate Editor F. Díaz-De-María. (*Corresponding author: P. Y. Mok.*)

The authors are with the School of Fashion and Textiles, The Hong Kong Polytechnic University, Hong Kong, and also with the Laboratory for Artificial Intelligence in Design, Innovation and Technology Fund, Hong Kong Science Park, Hong Kong (e-mail: tracy.mok@polyu.edu.hk).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2022.3213270>.

Digital Object Identifier 10.1109/TCSVT.2022.3213270

1051-8215 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See <https://www.ieee.org/publications/rights/index.html> for more information.

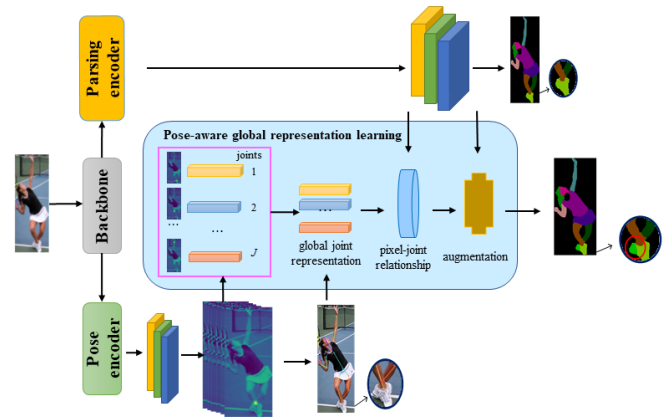


Fig. 1. Overview of the PAGRnet model.

The task of human parsing can be regarded as a form of semantic segmentation applied specifically to humans, and can therefore benefit from the advancements in general semantic segmentation research.

In recent years, the exploitation of richer contexts has been regarded as one of the most effective approaches to improving the performance of semantic segmentation [9], [10]. For instance, early attempts focused on introducing dilated convolution [9] or pyramid pooling operation [10] to cover a wider spatial context; however, these strategies failed to consider the class-level context. Following this, other approaches were reported including channel-wise [11], [12] or pixel-wise [13], [14], [15] global context learning. Although these proposals improved the performance of semantic segmentation, they tend to fail in human parsing, especially when context clues are not distinguishable from related labels. For example, when the labels of “left shoe” and “right shoe” become mixed up, the confused labels cannot be rectified using global information of the surrounding pixels.

Human parsing is distinguished from general semantic segmentation problems in terms of its unique focus and characteristics. More specifically, the target to be parsed (i.e. the human body) has a fixed skeleton, and the parser can learn more effectively if such information is leveraged. Pose estimation, which predicts the location of key human joints from input human images [16], provides effective information on the human body structure that can guide and refine the results of parsing. Some attempts have been made to combine the two tasks of human parsing and pose estimation to boost

the performance of each independent task [1], [17], [18], [19], [20]. However, these studies combined the pose estimation and human parsing tasks into a multi-task learning problem simply by fusing the intermediate features. In other words, these methods have exploited only the relationship between channel maps or pixels, and few of them have considered capturing the global context features of pose estimation to improve the results of parsing.

The use of multiple knowledge representation, namely combining multiple sources of representation, can improve the representation generality ability [21]. Yet, the best way to combine different representations is still an open question. We believe that integrating context information from pose estimation in human parsing at the pixel level has not been well explored in previous attempts, because no *class relationship* was used as context information. To illustrate our idea, we visualise the pairwise class relationship between parsing and pose labels in Fig. 2, in which the relationships are shown as “relevance” values in descending order. The relevance is obtained by calculating the numbers of overlapping pixels between the parsing and pose regions on the corresponding heatmaps, and is then normalised through division by the numbers of pixels in the corresponding parsing region. We calculated and visualised the *class relationship* in Fig. 2 using the entire validation set of 10,000 human images from the LIP dataset [1]. As shown, among 300+ pairwise class correlations, only around 15% (50 pairs) has relevance level over 0.3, and the rest has little relevance. Among the high relevant pairs, we can observe that some parsing labels have strong associations with specific body joints; for example, the relevance values between the left shoe region and the left ankle joint as well as between right shoe region and the right angle joints are both over 0.85, the highest among all pairs. For another example, the relevance values are 0.63 and 0.55 between the face region and the head joint and between the face region and the neck joint, respectively. There are also high associations between clothing regions and specific joints; for example, between upper clothing region and the spine joint as well as between dress and pelvis joint. Inspired by the fact that human joints are likely overlapped with different clothing/body regions on human images, we propose using joint representations to infer regions of clothing/body. If the position and appearance feature for areas near human joints are known, the representation of each parsing class can be extracted as the combination of relevant joint representations. Since joint representations consider the global feature near the joint, they are more stable and less redundant than pixel representations [1], [18], [19], [22], thus will benefit the parsing task.

In this paper, we propose a scheme that integrates global information embedded in human skeleton into the human parsing network (see Fig. 1). Specifically, our method exploits the *class relationship* between human parsing and pose estimation to improve parsing performance. Since human parsing is a pixel-level task, our parser learns the relationship between the pixel representation and the *global* joint representation to ensure the detailed local features are retained for human parsing. For the extraction of joint representation, we first

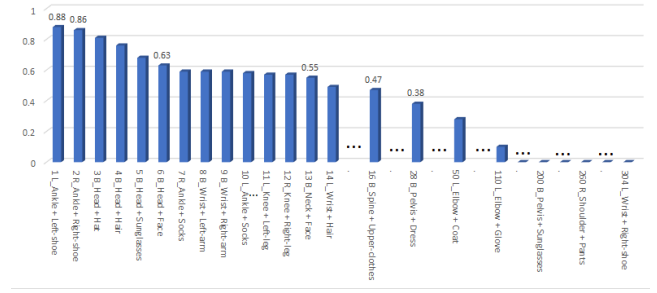


Fig. 2. Relevance between parsing and pose classes.

learn global information from pose/joint features using a global representation learning sub-module. Next, a relationship learning sub-module is applied to learn the relationship between the pixel representation and global joint representation. Finally, based on the global joint representation and its relationship with the pixel representation of human parsing, a new representation that includes pose-aware global information is obtained by augmentation for the human parsing task. A detail comparison between our PAGRnet with the pixel-wise formulation of CorrPM [19] will be given in later Sections III-D and IV-B.

We can summarise the key contributions of this paper as follows:

- (i) Different from all other fusion strategies of pose and parsing, we are the first, to the best of our knowledge, to use *global* joint feature of pose estimation to improve parsing, via a simple yet effective scheme of pose-aware global representation learning.
- (ii) We exploit the relationship between parsing pixel representation and the *global* representation embedded in human pose, and augment it to develop an enhanced representation to give better human parsing results.
- (iii) Our model improves the task of human parsing only with pose information, without the need for other information, e.g., boundaries. We evaluated our approach on three benchmark datasets, and found that it outperformed other methods that use pose for parsing, such as CorrPM [19], in terms of accuracy and efficiency. Our approach achieved competitive performance on benchmark datasets, for example, with mean Intersection-Over-Union (IoU) accuracy of 59.85% on the LIP [1], of 71.79% on the Pascal-Person-Part [23], and 86.21% of F1-score on the ATR datasets.
- (iv) Our proposed module of pose-aware global representation learning has an adaptable and flexible structure; it can work with different backbones and be deployed without precise pose information.

II. RELATED WORK

A. Semantic Segmentation

The task of semantic segmentation is to assign a label to each pixel for any given input image. Semantic segmentation has always been an active research topic in computer vision for many years. The remarkable improvements resulting from

the use of Full Convolutional Networks (FCNs) [24] have given rise to various FCN-based methods, which have become mainstream for use in semantic segmentation tasks. Since consecutive spatial pooling and convolutional strides reduce the resolution of the final feature maps, meaning that the feature maps lose much of their detail in terms of local information, some studies focused on recovering detailed local information and generating high-resolution features. A typical solution is to adopt an encoder-decoder structure that extracts better context information by downsampling using an encoder path, and then recovers the resolution to preserve spatial information using a decoder path. Representative methods include SegNet [25] and RefineNet [26]. To preserve the spatial information required for pixel label learning, FCN replaces the fully connected layers in the classification network with convolutional layers to provide a local receptive field. To enlarge the receptive field and better handle objects at multiple scales, DeepLab [27] designs the atrous convolutional kernel so as to capture more contextual information; based on this, DeepLab-v2 [9] and DeepLab-v3 [28] introduced ASPP, which used parallel dilated convolutional layers with different rates. Different from ASPP which fixes all the dilation rates, GDPP [29] dynamically sampled the middle dilation rates by the Gaussian distribution. PSPNet [10] uses a pyramid pooling module to combine the local and global information by aggregating different level features for more reliable prediction. SFANet [30] proposes a stage-aware feature alignment module (SFA) to address misalignment problem between different levels of feature maps during feature aggregation. Other studies focused on the adaptive fusion of different context information by learning contextual dependencies. For example, PSANet [31] learns the relationship (similarity) between pixels based on self-attention [32]. PCANet [33] fuses both channel and spatial attention mechanisms to capture long-range dependency to learn richer contextual features.

B. Human Parsing

Human parsing segments a human image into different body and clothing parts, which is a kind of fine-grained semantic segmentation. Generally speaking, semantic segmentation methods are applicable for human parsing. For example, Ruan et al. [34] showed that a PSP module, high-resolution decoder and edge-perceiving module are effective structures that could improve parsing performance. However, unlike semantic segmentation, human parsing involves the segmentation of the human body and clothing regions, and researchers in this field therefore focus more closely on how to exploit the known human skeleton and clothing information to improve segmentation performance.

Since the human body is highly structural, some researchers have modelled the context correlations between body parts using convolutional neural networks (CNNs) [35], recurrent networks with long short term memory (LSTM) units [23], or Macro-Micro Adversarial Networks [36]. Ji et al. [37] proposed semantic locality-preserving segmentation model which attaches a similar appearance or pose auxiliary exemplar to input image. Huang et al. [38] designed a pre-bilateral awareness block and a combined-order statistics awareness

block to balance feature mapping at trilateral levels so as to address the scale and inter-class distance imbalance problems in human parsing. Zhang et al. [39] proposed a Pyramidal Gather-Excite Context (PGEC) module to explore local and global clues by multiple GE operations with different spatial extents. Chan et al. [40] applied human detection to improve segmentation with changing environments. However, these methods did not exploit the strong hierarchical relationships between human body parts. Zhu et al. [41] developed a progressive cognitive network to recognise different body parts gradually based on a component-aware region convolution structure. Zhou et al. [42] proposed a Part-based Human Parsing Cascade (PHPC) to segment human images, based on observational mechanism of human being. Wang et al. [43] proposed to reason the human body parts to refine parsing results based on a hierarchical human body structure. Similarly, Li et al. [44] improved human parsing by a dual graph reasoning framework.

C. Human Pose for Parsing

Pose estimation predicts the coordinates of human joints from an image, which provides rich information on the structure of the human body. Although the aims of human parsing and pose estimation are different, both tasks involve understanding the human body in the input images. The two tasks are therefore likely complement one another. Human parsing focuses on an understanding of pixel-level information, and thus lacks global spatial information about the entire human body, which can yet be provided by pose estimation. In early studies, the segmentation of clothing/body is predicted given estimates of joint locations [3], [17], [18]. These methods usually train models separately for specific tasks, and only use the learned information for post-processing. This approach has a known drawback in that it is not able to optimise both pose and parsing tasks in the training stage.

Recently, some researchers attempted to jointly train both tasks in a network from a multi-task learning perspective [1], [19], [20], [22], [45]. Liang et al. [1] applied a refinement network to fuse the intermediate features via concatenation. Nie et al. [45] proposed a mutual adaptation module to connect the learning of pose and parsing. Lin et al. [20] used human pose estimation as an auxiliary task to assist human part segmentation without human-annotated segmentation labels. However, these methods were not able to exploit the relationship between human segmentation and pose estimation. Zhang et al. [19] exploited the relationships between parsing pixels and pose pixels, and used the calculated relationships as weights to enhance the parsing feature as part of a self-attention scheme [32]. In this work [19], pose was represented by pixel features; individual joint pixels may not give clear clues to the global information about the human body. Zeng et al. [22] proposed to use Neural Architecture Search (NAS) to search encoder and decoder architectures for human parsing and pose estimation, as well as the feature fusion structure to improve both tasks. Yang et al. [46] pretrained the pose network to first obtain pose prediction and then transferred the pose knowledge to human parsing network by knowledge distillation.

Hence, the problem described in the Introduction, in relation to the example of confusion between the left and the right shoes, still remains unsolved. ***It is important to determine the best way to aggregate global context information.***

D. Global Context Learning

The integration of global context are known to be an effective way to improve the task of semantic segmentation. A typical method of generating global representation is to aggregate features from all pixels via global average pooling. BiSeNet [47] exploits global pooling for a large receptive field and uses the global context as a attention to guide the feature learning. DFN [48] adds global average pooling at the end of the backbone network to compute the global context, which are used to guide the feature learning of lower stages. However, the global pooling operation sums up the feature of all pixels with the same weighting.

In contrast, OCNet [49] learns a relation matrix between pixels as the inter-pixel similarity, and compute a pixel representation by multiplying the representation of each pixel with its inter-pixel similarity to this pixel and summing the features of all pixels. DANet [11] applies not only the relationship between pixels but also the relationship between channels to calculate the representation of each pixel by traditional self-attention module, which is computational expensive. CCNet [19] employs criss-cross attention to reduce the computation cost. In addition, ACFNet [13] calculates the average of the pixel representations for all pixels belonging to each class, called the class center, and then exploits the relations between classes to update the pixel representation. DGCWNet [12] learns the relationship between pixels for each channel and update the channel representation by aggregating representation of all pixels. OCRNet [14] explores the relation between pixel representation and class-level representation, and uses the class-level representation to represent a pixel-level representation for each pixel. Fan et al. [15] exploited the class-level representation to improve the performance of pixel-level prediction in video object segmentation.

In contrast, our approach generates the global representation *from another related task* rather than the target task itself, and uses a global information learning module to learn the relationship between the global representation captured from the related task and the pixel representation of the target task. ***Our proposed learning module connects two different tasks together, it therefore allows the network to capture the commonalities and differences among tasks.***

III. PROPOSED METHOD

In this section, we introduce a novel pose-aware global representation network model, denoted as PAGRnet, for human parsing, based on the integration of global information learned from pose estimation (see Fig. 1 for an overview and Fig. 3 for detail network structure based on ResNet-101 backbone). We present the overall framework in Section III-A, and then the parsing and pose feature encoders in Section III-B. In Section III-C, we explain the pose-aware global representation learning module in detail, which is compared to other

pixel-wise feature fusion scheme in Section III-D. Lastly, the overall training objective is given in Section III-E. Our codes are shared at <https://github.com/TracyMok/PAGRnet.git>

A. Problem Formulation

Given an input image $I \in \mathbb{R}^{3 \times (\mu \times \lambda)}$ of size $\mu \times \lambda$, our goal is to predict the label, $S_i \in \{0, 1, \dots, L\}_i^{\mu \times \lambda}$, for each pixel i , based on the estimated locations of the human body joints $P = (u_j, v_j)_{j=1}^J$. Here (u_j, v_j) denote the coordinates of the j^{th} joint, J denotes the total number of body joints, and the human parsing task has a total of $L + 1$ classes.

As human parsing and pose estimation are complementary and closely related, they can be trained jointly by multi-task learning. In our network, we use a backbone model to share the lower-level features, a parse encoder and a pose encoder to learn the specific feature representations of parsing and pose, denoted as $f_s \in \mathbb{R}^{c \times h \times w}$ and $f_p \in \mathbb{R}^{c \times h \times w}$, respectively.

As shown in the detail network structure based on ResNet-101 backbone of Fig. 3, our scheme first generates the global representation $g_p \in \mathbb{R}^{J \times c}$ from the pose feature f_p using a *global representation generation sub-module*, and then learn the relation between f_s and g_p using a *pixel-joint relationship sub-module*. Finally, the pose-aware global representation for each pixel i can be computed by:

$$y_i = \rho\left(\sum_{j=1}^J \omega_{ij} \delta(g_{pj})\right) \quad (1)$$

where ω_{ij} is the relation between f_{si} and g_{pj} ; $\delta(\cdot)$ and $\rho(\cdot)$ are transformation functions. Such global representation y_i is augmented to the original parsing feature f_{si} in a *relation augmentation sub-module* for the final refined parsing prediction.

B. Feature Encoding

As shown in Fig. 3, feature encoding consists of a backbone (base model), a parsing encoder and a pose encoder.

1) *Backbone*: For the base model, we use ResNet-101 [50] pretrained on ImageNet [51]. Following CE2P [34], we remove the classification layer and the last two pooling layers, and set the dilation rate of the convolution layers succeeding the removed pooling layers to 2. In this way, the output size is reduced to one sixteenth (1/16) of the original image size. Convolutional features from all stages in the backbone network are used as inputs for the following encoders. More specifically, we choose the higher-stage features after “Res-5” layer of the base model for the pose encoder, since these features contain richer semantic information, which is helpful in capturing global semantic pose information. For the parsing encoder, we apply both low-level and high-level features, since low-level features contain more detailed information and can contribute to the task of parsing small items such as sunglasses and socks.

2) *Parsing Encoder*: Context information has been shown in many previous studies to be effective for both semantic segmentation and human parsing. We therefore add atrous spatial pyramid pooling (ASPP) [9] to enlarge the receptive field

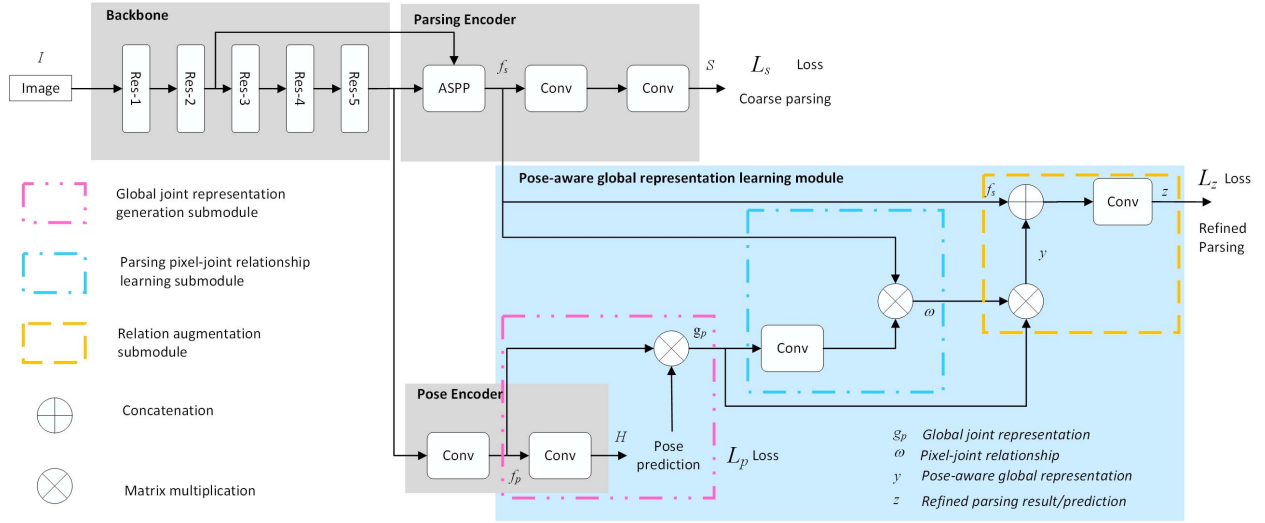


Fig. 3. Detailed network structure of the proposed PAGRnet model.

and acquire more powerful context information. ASPP consists of five parallel parts: one global average pooling branch, one 1×1 convolution branch, and three 3×3 convolution branches with dilation rates of 12, 24 and 36, respectively. Except for the last convolution, which has 512 channels, the number of channels for other convolutions is 256. To improve the segmentation of small items, following [28], we concatenate the output features of the base model and lower-stage features extracted from the “Res-2” layer of the base model. We then input the concatenated features into a 1×1 convolution and output the feature f_s of size $h \times w$ in $c = 512$ channels, where $h = \mu/4$ and $w = \lambda/4$. Finally, a 1×1 convolutional layer is used to generate the coarse parsing results of our model. The pixel-wise output prediction of S has one fourth of original image size I , which is scaled back to the size of original ground-truth labels for loss calculation.

3) *Pose Encoder*: This module generates the estimated locations of human body joints $P = (u_j, v_j)_{j=1}^J$ and captures the global representation of the human body. Since human parsing is a pixel-level task, we formulate the pose estimation as a pixel regression problem, similar to that in [52], to retain the location consistency of the pose and parsing features. There are many approaches to pose estimation in which complicated CNNs are used to obtain more accurate key point locations, but these involve high computational cost. To balance between efficiency and accuracy, we apply a simple but effective network [53] for pose estimation. More specifically, we add two transposed convolution layers based on the base model to generate the pose feature representation f_p , and then regress the heatmap from it. Following [54], we take the joint coordinate as the center and generate a 2D Gaussian probability distribution map with a standard deviation of seven pixels, and then add a 1×1 convolution layer to generate the pose prediction heatmaps $H = \{H_j\}_{j=1}^J$, $H_j \in \mathbb{R}^{h \times w}$.

C. Pose-Aware Global Representation Learning

1) *Global Joint Representation Generation*: Global context has proven to be effective in terms of improving the performance of semantic segmentation [11], [13], [14].

Instead of generating a global context representation from all pixels of each class, we attempt to generate a global representation from the related task of pose estimation. To achieve this, we first normalise the estimated heatmaps $H = \{H_j\}_{j=1}^J$ to $M = \{M_j\}_{j=1}^J$. Next, the pose feature representation from pose encoder is multiplied by the normalised probability of human joint j as weight at each pixel level. Then, we sum up the representations of all pixels to obtain g_{pj} , the global representation of joint j , by:

$$g_{pj} = \sum_i M_{ji} f_{pi} \quad (2)$$

where f_{pi} is the pose representation of pixel i , and M_{ji} is the normalised probability of the j^{th} human joint occurring at pixel i .

The benefit of global joint representation (g_{pj}) is that it allows each pixel to understand the overall representation of a joint from a global view. Since global joint representation aggregates the pixel representations within the joint’s neighborhood, it contains more global and accurate features of the joint than any single pixel, and therefore g_{pj} is more beneficial to enhance the pixel representation near the corresponding joint positions.

2) *Parsing Pixel-Joint Representation Relationship Learning*: To effectively enhance all the pixel representations for human parsing, we propose a parsing pixel-joint representation learning module to learn the relationship $\omega_{ij} \in \mathbb{R}^{h \times w \times J}$ between the pixel representation of the parsing feature f_{si} and the global representation of the pose feature g_p as follows:

$$\omega_{ij} = \frac{e^{\kappa(f_{si}, g_{pj})}}{\sum_{j=1}^J e^{\kappa(f_{si}, g_{pj})}} \quad (3)$$

As shown, we compute the relationship ω_{ij} between each pixel i and each joint j . In (3), $\kappa(f_s, g_p) = f_s^T \psi(g_p)$, where $\psi()$ is transformation function consist of a set of convolutions followed by batch normalization and ReLU. Therefore, our formulation allows the network to learn global information with reference to all joints J for the parsing task.

3) *Relation Augmentation*: By (1), we can calculate the pose-aware global representation y_i for each pixel i using the generated g_p and the learned ω_{ij} . The final parsing prediction z_i is updated by integrating the original pixel representation of parsing feature f_{si} with the pose-aware global representation y_i as follows:

$$z_i = \phi([f_{si}, y_i]) \quad (4)$$

where $[]$ is a concatenation operation, and $\phi()$ is the transformation function implemented via a 1×1 convolution followed by batch normalization, dropout operation, and one more 1×1 convolution.

We use the learned pixel-joint relation to augment global pose representation for the task of human parsing, to take advantage of the correlation between parsing classes and human joints. With this correlation, assuming the parsing class of a particular pixel is known, we can project how this pixel corresponds to different joints. For example, if strong correlations are found between upper-clothing and the right-shoulder, left-shoulder, spine and pelvis joints, then all the pixels of a upper-clothing should correspond to these joints. By updating the pixel representation with this relation, the pixel representation can be enhanced based on the learned relation between human joints and parsing classes.

D. Global Representation vs. Pixel-Wise Feature Fusion

In this section, we compare the formulations of feature representation between our PAGRnet and that of CorrPM [19]. As illustrated in Fig. 4(a), CorrPM [19] calculated the pixel-pixel relationship between the parsing features and pose features, and used this pixel-wise relationship to enhance parsing learning. The pixel-wise feature fusion scheme and our formulation (shown in Fig. 4(b)) have threefold differences as follows.

Firstly, the network structure of [19] takes the original pixel representation of pose f_p and focuses on the correlation between it and parsing feature f_s , while PAGRnet takes the global pose representation g_p and focuses on the correlation between it and parsing feature. According to (2), each heatmap H_j is multiplied with the original pose feature f_p and then summed up over all pixels to generate the global representation g_p . Since each joint heatmap focuses on the neighbourhood pixels around the joint, g_p has a wider field of vision and with more “global” information.

Secondly, the two formulations capture different correlations. CorrPM [19] captures the spatial dependencies between any two pixels of the feature maps. With reference to Fig. 4(b), original pose feature f_p and parsing feature f_s are first concatenated together and then fused by a convolution to generate feature f_h . After that, a self attention is applied to f_h to learn the pixel-wise relation map $S \in \mathbb{R}^{hw \times hw}$. The more similar the representations of the two pixels, greater the correlation between them. PAGRnet calculates the relation ω_{ij} between each pixel representation f_{si} and each joint representation g_{pj} by the dot product (as seen in (3)), and augment the representation of each pixel with the global joint representation of all joints. High response value of w represent

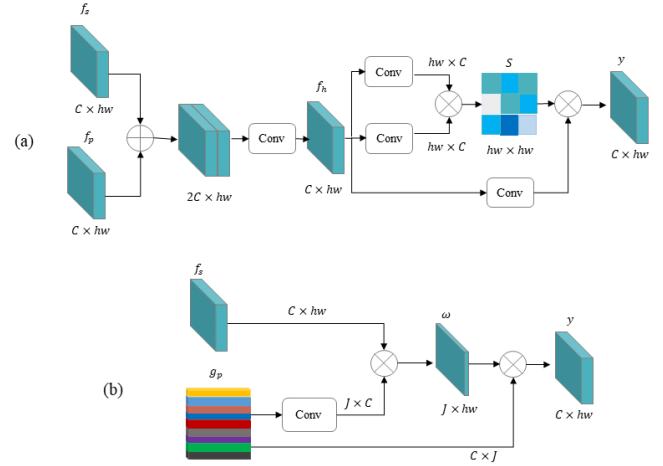


Fig. 4. Comparison of network structure for feature fusion module: (a) pixel-wise feature fusion [19] (b) our proposed pose-aware global representation.

high correlations with the corresponding joints. Therefore, the proposed pose-aware global representation learning module implicitly learns the *class relationship* between human joints and parsing classes. In sum, the learning of human parsing can be strengthened by this relation map in accordance with the correlations between the parsing class and different joints.

Thirdly, PAGRnet has lower computation complexity compared with pixel-wise formulation while obtains higher accuracy. As shown in Fig. 4, when calculating the correlation matrix PAGRnet only applies a convolution to transform the global joint representation (g_p) and the correlation matrix is performed by a matrix multiplication between the feature maps $\mathbb{R}^{J \times C}$ and the feature maps $\mathbb{R}^{C \times hw}$. The computation complexity is $\mathcal{O}(J \cdot hw \cdot C)$. The pixel-wise feature fusion applies two convolutions on $f_h \in \mathbb{R}^{C \times hw}$ and a convolution on $f_p \in \mathbb{R}^{C \times hw}$, and the correlation matrix is performed by a matrix multiplication between the feature maps $\mathbb{R}^{hw \times C}$ and the feature maps $\mathbb{R}^{C \times hw}$. The whole computation complexity is $\mathcal{O}(hw^2 \cdot C)$. We will present a comparative study on our PAGRnet and pixel-wise feature fusion schemes in later Section IV-B.

E. Loss Function

We not only use the parsing loss but also the pose loss to supervise the network training. The overall training loss function is defined as:

$$L = \alpha_z L_z + \alpha_s L_s + \alpha_p L_p \quad (5)$$

where the parsing loss L_z is the loss between the parsing results Z and the parsing annotations, L_s is an auxiliary parsing loss denoting the loss for the coarse parsing results S , the pose loss L_p is the loss between the joint locations prediction H and the true joint locations, and α_z , α_s and α_p are weight parameters for parsing loss L_z , auxiliary parsing loss L_s , and pose loss L_p , respectively. We compute cross entropy loss for L_s and L_z . Heatmaps of the joint locations are beneficial to preserve spatial information and are widely used in pose estimation. We therefore encode the joint locations

as heatmaps using the 2D Gaussian probability density function [52] and compute the mean square error as pose loss L_p .

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

To evaluate the effectiveness of our proposed method, we conducted comprehensive experiments on three benchmark datasets. In the following subsections, we describe these datasets, the evaluation metrics, the details of our implementation and the results in comparison to other methods. We performed ablation study on the LIP dataset in Section IV-C. Finally, we also report the results on the Pascal-Person-Part and the ATR datasets.

A. Datasets, Metrics and Experimental Details

1) *The LIP Dataset*: [1] is a large-scale benchmark dataset focusing on the fine-grained segmentation of human body parts and clothing. It contains 50,462 images, each of which has annotated location coordinates of 16 key points of the body joints ($J = 16$) and pixel-level labels of 20 human/clothing parts ($L = 19$). The dataset is partitioned into training set (30,462 images), validation set (10,000 images) and test set (10,000 images).

2) *The Pascal-Person-Part Dataset*: [23] is a dataset with 3,533 multiple-person images, each with annotations representing human part segmentation labels of the head, torso, upper arms, lower arms, upper legs, and lower legs ($L = 6$). The dataset is partitioned into training set of 1,716 images and testing set of 1,817 images. We use “Detectron2” [55] to first segment the mask of every single person and then obtain 17 key point locations ($J = 17$) for the segmented person as ground truths. By so doing, we obtain 4,303 single-person images for training. For testing, we first segment each image in the same way, then input the cropped single-person images to the trained model for parsing prediction, and finally combine parsing results for all persons based on the original image for result evaluation.

3) *The ATR Dataset*: [35] is a human parsing dataset of fashion images. There are a total of 7,700 images annotated with 18 human/clothing parts ($L = 17$) similar to the LIP dataset [1]. The dataset is randomly split into 6,000 images for training, 1,000 for testing and 700 for validation. Another dataset Chictopia10k [56] is also added into the training set to increase data diversity, like [35], [57], [58]. Similar to that of Pascal-Person-Part dataset, we use the pose estimator of [55] to obtain 17 key point locations ($J = 17$) as the pose ground truths.

4) *Evaluation Metrics*: To facilitate comparative analyses, we used the same set of evaluation metrics of the FCN in our experiments on the LIP dataset [1] and the Pascal-Person-Part dataset [23]. We scaled the output pixel-wise prediction back to the size of the original ground-truth labels, and calculated the pixel accuracy, mean accuracy, and mean accuracy on the intersection over the union region (IoU) for evaluation. For the ATR dataset [35], we calculated the pixel accuracy, foreground accuracy, average precision, average recall and average F-1 score for comparison.

5) *Implementation Details*: The size of the input images was 384×384 on all the three datasets. The stochastic Gradient Descent (SGD) strategy was used to optimize the network with ResNet-101 backbone in all experiments with the following parameters: the initial learning rate of 0.005, the weight decay of 0.0005, and the momentum of 0.9. We adopted the poly policy [28], [59] in the training process, in which the learning rate is updated after each iteration by multiplying $(1 - \frac{t}{t_{\text{total}}})^{0.9}$, where t is the iteration number and t_{total} is the total number of iterations. For a fair performance comparison on networks with ResNet-101 backbone, following [19], we set the loss weights in (5) as $\alpha_z = \alpha_s = 1$ and $\alpha_p = 70$.

For networks with HRNetV2-W48 backbone, we trained using Adam optimizer with an initial learning rate of 0.001 and dropped by one-tenth at epoch 170. The loss weights were set as $\alpha_z = 0.4$, $\alpha_s = 1$ and $\alpha_p = 1000$, respectively. Since the learning of L_s depends on the learning of relationship between parsing representation and global pose representation, which is difficult and needs more attention, thus the weight for L_z is set as larger than that for L_s . We set a large weight for pose estimation loss L_p because our method exploits pose estimation for a good global representation.

To avoid overfitting and improve network robustness, we applied some common data augmentation strategies in the training, including random cropping, left-flipping, and random scaling from 0.75 to 1.25. All batchnorm layers in the network were replaced by INPLACE-ABN [60] and the batch size was set to eight. We fine-tuned the network using the model pre-trained on the ImageNet for 180 epochs, and we conducted our training and experiments using two NVIDIA 1080Ti GPUs for all results reported later in Tables I-VI.

B. Evaluation on the LIP Dataset

In Table I, we compare our method with other reported work that also used pose estimation for improving parsing (as discussed in Section II-C). Since MuLA [45] requires the input size to be 256×256 , we also trained PAGRnet using the same input size for a fair comparison. As shown in Table I, our method outperforms MuLA [45] by 1.96% of average mean IoU. Compared to [1] which used input size 384×384 , our method outperforms 4.48% in terms of mean IoU. CorrPM [19] uses both pose and edge information to improve parsing. To explore the effect of pose information for a fair comparison, we trained their model fusing with only pose features (see Fig. 4(a)) for the same 180 epochs. As listed in Table I, our method outperforms CorrPM [19] by 0.85%. Compared to SMDC [46], which also uses pose and edge to improve human parsing, our method outperforms by 0.64% in terms of mean IoU.

We also compare in Table I the efficiency of most these methods in terms of computational complexity (FLOPs, namely floating-point operations [61]) and the number of parameters. The efficiency performance of SMDC [46] are missing in Table I, because the authors did not provide such comparison or the detail network configuration in their article [46]. As shown in Table I, our method is superior to [1] in terms of both model parameters and FLOPs, because [1]

TABLE I
COMPARISON OF RESULTS: ACCURACY AND EFFICIENCY ON THE VALIDATION SET OF LIP DATASET

Methods	Input size	Mean IoU	Parameters	FLOPs
MuLA [45] [†]	256×256	49.30%	42.6M	42.6G
JPPNet [1]	384×384	51.37%	79.2M	171.3G
CorrPM [19] ^{§, †}	384×384	55.00%	70.3M	87.7G
SMDC [46] ^{§, †}	473×473	55.21%	-	-
NPPNet [22] [†]	384×384	57.14%	73.4M	113.7G
PAGRnet-ResNet101 (ours) [†]	256×256	51.26%	70.0M	37.0G
PAGRnet-ResNet101 (ours) [†]	384×384	55.85%	70.0M	83.2G
PAGRnet-HRNetV2-W48 (ours) [†]	384×384	58.39%	73.0M	108.3G

[†] means single-scale testing results without flipping;

[§] refers to CorrPM [19] result of fusing parsing features with only one pose features.

TABLE II
ABLATION STUDY ON THE LIP DATASET: PER-CLASS AND AVERAGE MEAN IOU (%) [†]

Methods	Hat	Hair	Glove	Glass	U-cloth	Dress	Coat	Sock	Pants	Jumpsuit	Scarf
baseline-ResNet101	65.14	70.95	41.36	31.11	68.80	36.12	55.63	47.50	74.13	30.77	19.13
concat-ResNet101 [f_s, f_p]	66.18	71.41	41.38	31.36	69.34	35.38	55.36	48.11	74.88	31.80	21.43
global-ResNet101 [f_s, g_p]	66.30	71.09	42.16	30.43	69.70	36.79	56.91	47.15	74.96	28.81	21.53
PAGRnet-ResNet101 [f_s, y]	66.77	71.54	42.22	31.35	70.23	39.25	57.06	50.87	75.10	30.18	23.84
CorrPM [19] (local-ResNet101)	65.70	71.71	42.36	31.13	70.00	36.08	57.20	48.30	75.02	32.73	21.88
baseline-HRNetV2-W48	65.50	71.64	42.44	37.04	67.81	28.80	53.42	49.53	73.93	24.11	24.26
concat-HRNetV2-W48	69.18	73.13	47.03	39.87	69.42	35.21	55.65	49.53	76.09	27.49	26.43
PAGRnet-HRNetV2-W48	68.06	72.11	49.94	37.79	69.84	39.91	57.11	51.69	76.38	28.33	27.07
Methods	Skirt	Face	L-arm	R-arm	L-leg	R-leg	L-shoe	R-shoe	Bkg	Ave.	
baseline-ResNet101	28.09	74.01	64.21	66.76	59.95	59.00	45.47	45.51	87.26	53.55	
concat-ResNet101 [f_s, f_p]	28.78	74.09	65.76	67.88	61.77	61.82	48.27	49.46	87.58	54.59	
global-ResNet101 [f_s, g_p]	29.70	74.55	66.82	69.06	64.26	63.05	50.45	51.35	87.66	55.13	
PAGRnet-ResNet101 [f_s, y]	31.25	74.52	67.17	69.23	63.91	64.00	50.09	50.65	87.83	55.85	
CorrPM [19] (local-ResNet101)	28.24	74.60	66.44	68.52	62.00	61.23	49.01	50.08	87.73	55.00	
baseline-HRNetV2-W48	22.27	75.06	66.41	68.35	64.88	63.60	52.38	52.71	87.40	54.58	
concat-HRNetV2-W48	27.76	75.94	69.60	71.80	67.27	67.50	56.26	56.19	88.24	57.53	
PAGRnet-HRNetV2-W48	29.81	75.79	70.06	71.84	70.22	68.92	57.24	57.68	88.06	58.39	

[†] The results shown in this table are all single-scale testing results without flipping. | **bold** – the highest value.

used three sub-networks to refine the parsing results. Although the number of parameters of our method with ResNet101 backbone is more than that of MuLA [45], our method has lower FLOPs.

As discussed in Section III-D, CorrPM [19] uses pixel-wise feature fusion, our method with ResNet-101 backbone has fewer parameters (70.3M vs. 70.0M) and lower FLOPs (87.7G vs. 83.2G), because we exploit the global representation with lower dimensions ($f_p \in \mathbb{R}^{C \times h \times w}$ vs. $g_p \in \mathbb{R}^{J \times h \times w}$) and directly calculated the *relation* w between parsing feature and transformed global pose representation, which leads to lower time complexity ($\mathcal{O}((hw)^2 \cdot C)$ vs $\mathcal{O}(J \cdot hw \cdot C)$). Our method PAGRnet outperforms CorrPM [19] by 0.85% with fewer parameters and FLOPs. Although the improvement seems not high, the improvement of human parsing is difficult especially without increasing much parameters. For example, CNIF [43] is 0.69% better than PCNet [62] in terms of average mean IoU, but increases 32.5M parameters and 64.6G FLOPs. HHP [63] is 1.51% better than CNIF [43], but the processing speed is slowed down by half. Compared to NPPNet [22], our method with HRNetV2-W48 backbone performs better in terms of Mean IoU (57.14% vs 58.39%), with fewer parameters (73.4M vs 73.0M) and FLOPs (113.7G vs 108.3G). With ResNet-50 backbone, NPPNet uses two independent networks for human

parsing and pose estimation, and adds the feature interaction between multi-scale features. On the contrary, we share the backbone parameters of human parsing and pose estimation and fuse the global representation and parsing representation to reduce the computation as far as possible. Therefore, although we use a larger backbone of HRNetV2-W48, our model has fewer parameters and FLOPs than NPPNet [22]. This demonstrates that our method effectively improves the task of human parsing using pose information without substantially increasing the complexity of the model.

C. Ablation Studies

To evaluate the effectiveness of our *pose-aware global representation learning module* covering three sub-modules (as illustrated by the blue region in Fig. 3), we compare the performance of the baseline model that without the use of pose estimation for parsing (denoted as “base”), a model that simply concatenates pose and parsing features (denoted as “concat” [f_s, f_p]), a model that concatenates upsampled global joint representation and parsing features (denoted as “global” [f_s, g_p]), and our PAGRnet model ([f_s, y]) on the LIP dataset in Table II. The results of model with pixel-wise feature fusion of [19] (denoted as “local”) is also shown in the

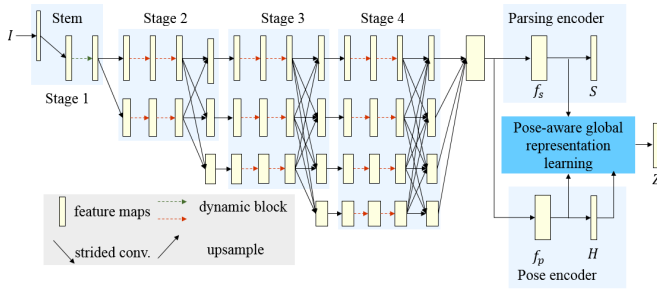


Fig. 5. PAGRnet network structure on HRNetV2-W48 baseline.

same table for easy comparison, in which the highest mean IoUs are highlighted in bold.

1) *Effectiveness of the Global Representation g_p* : To verify the effectiveness of g_p , (as illustrated by the dotted pink region in Fig. 3), we first remove the parsing pixel-joint relationship learning sub-module (the dotted blue region in Fig. 3). The generated global joint representation g_p is upsampled to $\mathbb{R}^{512 \times h \times w}$ and then concatenated with parsing feature f_s to give the final parsing results. As shown in Table II, compared with the baseline model, the concatenation of the original pose feature f_p and parsing feature f_s has improved the average mean IoU by 1.04% (i.e. 54.59%–53.55%). Compared with simple concatenation, the use of global joint representation g_p has further improved the performance from 54.59% to 55.13% using baseline of ResNet101. For the per-class performance, the mean IoU value has improved in most classes, except for a few classes that are weakly associated with human joints. This is because global joint representation has less information about pixels of these classes. Nevertheless, for classes that are closely related to human joints, the use of global joint representation g_p has yielded large margins of improvement in mean IoU, 2.49% for the left leg, 1.23% for the right leg, 2.18% for the left shoe and 1.89% for the right shoe. This has illustrated the effectiveness of the generated global joint representation g_p .

2) *Effectiveness of Augmented Joint Global Representation*: We further evaluate the role of augmented joint global representation using parsing pixel-joint relationship learning sub-module. From the experimental results shown in Table II, compared with the experiment of concatenating global joint representation, the parsing pixel-joint relationship learning sub-module has further improved the performance from 55.13% to 55.85%. On the other hand, our proposed pose-aware global representation outperforms the previous “local” method [19] of pixel-wise feature fusion (Fig. 4(a)) by 0.85% (i.e. 55.85%–55.00%) on average mean IoU and achieved better performance on most labels in terms of per-class mean IoU. Specifically, our method performs better than the pixel-wise feature fusion method by 3.17% on Dress, 3.01% on Skirt, and 2.77% on R-leg, respectively. This indicates that our proposed pose-aware global representation is effective in terms of improving human parsing performance and offers good performance improvement for classes that are closely related to human joints, as shown in Fig. 2.

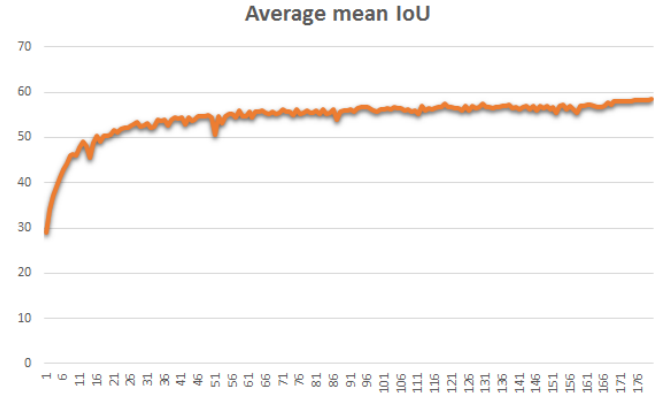


Fig. 6. The performance (average mean IoU) against training epochs of PAGRnet based on HRNetV2-W48.

3) *Flexible and Adaptable Structure*: We also train PAGRnet using other baseline model HRNetV2-W48 [64], which has excellent performance on both semantic segmentation [14] and pose estimation [54]. As shown in Fig. 5, we extract the four stages of parallel convolution streams as the backbone, and its output features are upsampled to the input size by bi-linear upsampling and input into the parsing and pose encoders. As HRNetV2-W48 [64] fully aggregates high-level and low-level representation by repeatedly connecting high-to-low convolution streams in parallel, we only used one convolution to encode the parsing feature and pose feature, respectively. Finally, the proposed pose-aware global representation learning module is used to fuse the parsing and pose features to generate the final parsing results. Fig 6 shows the performance (average mean IoU on the LIP validation dataset) against training epochs of our PAGRnet based on HRNetV2-W48.

As shown in Table II, the baseline model of HRNetV2-W48 obtains the mean IoU of 54.58%, which is improved to 58.39% by implementing our PAGRnet. As shown, the PAGRnet based on HRNetV2-W48 obtains larger improvement than that based on ResNet101, because the stronger backbone of HRNetV2-W48 not only predicts more precise semantic segmentation but also estimates more precise human pose [14], [54]. It is important to note that both ResNet101 and HRNet-48 have achieved reasonable improvements by introducing the proposed pose-aware global representation learning. The proposed module has an adaptable structure and can work flexibly with different backbones.

D. Comparison With State-of-the-art Methods on LIP Dataset

1) *Quantitative Comparison*: We compare our method with 16 major state-of-art methods on the LIP validation set, as shown in Tables III and IV. Some pre-class comparisons are missing in Table IV, e.g. [45], [63], because either these authors did not publish per-class results or such results were not reproducible from their shared models.

In addition to defeating all three methods using pose information, Table III also shows that our method, based on ResNet101, outperforms most other state-of-the-art methods,

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS IN TERMS OF OVERALL ACCURACY ON THE VALIDATION SET OF LIP DATASET

Methods	Input size	Edge	Pose	Pixel acc. (%)	Mean acc. (%)	Mean IoU (%)
MMAN (ECCV18) [36]	256×256			-	-	46.81
MuLA (ECCV18) [45] [†]	256×256		✓	88.50	60.50	49.30
SS-NAN (CVPRW17) [65]	321×321			87.59	56.03	47.92
JPPNet (PAMI18) [1]	384×384		✓	86.39	62.32	51.37
CE2P+Flip (AAAI19) [34]	473×473	✓		87.37	63.20	53.10
PGECNet (TCSVT21) [39]	473×473	✓		87.50	65.66	54.30
SLRS (CVPR20) [44] [†]	473×473	✓		88.33	66.53	56.34
BraidNet (ACMMM19) [66] [†]	384×384			87.60	66.09	54.42
SMDC (TC21) [46] [†]	473×473	✓	✓	88.12	66.53	55.21
CorrPM (CVPR20) [19] [†]	384×384	✓	✓	87.68	67.21	55.33
OCR (ECCV20) [14]	473×473			-	-	56.65
PCNet (CVPR20) [62]	473×473	✓		-	-	57.03
CNIF (ICCV19) [43]	473×473			88.03	68.80	57.74
NPPNet (ICCV21) [22]	384×384	✓	✓	-	-	58.56
HHP (CVPR20) [63]	473×473			89.05	70.58	59.25
SCHP (PAMI20) [58]	473×473			-	-	59.36
PAGRnet-ResNet101 (ours) [†]	384×384		✓	87.74	67.80	55.85
PAGRnet-ResNet101 (ours)	384×384		✓	88.23	68.32	57.06
PAGRnet-HRNet-W48 (ours) [†]	384×384		✓	88.01	72.53	58.39
PAGRnet-HRNet-W48 (ours)	384×384		✓	88.75	72.73	59.85

[†] means the single-scale testing results without flipping.

TABLE IV
COMPARISON WITH STATE-OF-THE-ART METHODS ON PER-CLASS MEAN IOU (%) ON THE VALIDATION SET OF LIP DATASET

Methods	Hat	Hair	Glove	Glass	U-cloth	Dress	Coat	Sock	Pants	Jumpsuit	Scarf
MMAN (ECCV18) [36]	57.66	65.63	30.07	20.02	64.15	28.39	51.98	41.46	71.03	23.61	9.65
SS-NAN (CVRPW17) [65]	63.86	70.12	30.63	23.92	70.27	33.51	56.75	40.18	72.19	27.68	16.98
JPPNet (PAMI18) [1]	63.55	70.20	36.16	23.48	68.15	31.42	55.65	44.56	72.19	28.39	18.76
CE2P (AAAI19) [34] †	65.29	72.54	39.09	32.73	69.46	32.52	56.28	49.67	74.11	27.23	14.19
PGECNet (TCSVT21) [39]	66.36	72.83	40.76	32.85	69.93	33.78	56.48	48.86	74.51	28.51	25.16
SLRS (CVPR20) [44]†	65.74	71.55	42.58	30.62	69.44	37.13	56.05	47.34	74.92	31.18	23.77
BraidNet (ACMMM19) [66] †	66.80	72.00	42.50	32.10	69.80	33.70	57.40	49.00	74.90	32.40	19.30
SMDC (TC21) [46]†	66.76	72.98	42.91	32.77	70.24	37.78	57.40	50.88	75.46	32.38	23.26
CorrPM (CVPR20) [19] †	66.20	71.56	41.06	31.09	70.20	37.74	57.95	48.40	75.19	32.37	23.79
OCR (ECCV20) [14] †	67.42	73.16	44.01	36.31	69.90	35.53	57.03	51.87	75.09	29.85	19.16
PCNet (CVPR20) [62]	69.32	73.08	44.72	34.21	72.59	36.02	60.84	51.03	76.66	38.78	31.60
CNIF (ICCV19) [43]	69.55	73.45	45.17	41.45	70.57	38.52	57.94	54.02	75.07	28.00	31.92
NPPNet(ICCV21) [22]	66.43	72.34	51.98	31.59	71.88	40.88	60.54	49.81	77.07	27.55	26.58
SCHP (PAMI20) [58]	70.63	74.09	51.40	41.70	70.56	40.06	58.17	55.17	76.57	33.78	26.63
PAGRnet-HRNetV2-W48 (ours) †	68.01	72.11	49.94	37.79	69.84	39.91	57.11	51.69	76.38	28.33	27.07
PAGRnet-HRNetV2-W48 (ours)	69.33	73.59	51.34	39.46	71.82	39.46	58.83	53.21	77.64	31.33	29.81
Methods	Skirt	Face	L-arm	R-arm	L-leg	R-leg	L-shoe	R-shoe	Bkg	Ave.	
MMAN (ECCV18) [36]	23.20	69.54	55.30	58.13	51.90	52.17	38.58	39.05	84.75	46.81	
SS-NAN (CVRPW17) [65]	26.41	75.33	55.24	58.93	44.01	41.87	29.15	32.64	88.67	47.92	
JPPNet (PAMI18) [1]	25.14	73.36	61.97	63.88	58.21	57.99	44.02	44.09	86.26	51.37	
CE2P (AAAI19) [34] †	22.51	75.5	65.14	66.59	60.10	58.59	46.63	46.12	87.67	53.10	
PGECNet (TCSVT21) [39]	26.52	75.34	65.69	67.33	59.36	58.82	47.71	47.78	87.74	54.30	
SLRS (CVPR20) [44] †	30.44	74.70	64.73	67.27	57.18	57.89	45.82	46.30	87.70	54.12	
BraidNet (ACMMM19) [66] †	27.20	74.90	65.50	67.90	60.20	59.60	47.40	47.90	88.00	54.40	
SMDC (TC21) [46]†	27.74	75.98	65.81	68.09	60.30	59.87	47.61	47.89	88.03	55.21	
CorrPM (CVPR20) [19] †	29.23	74.36	66.53	68.61	62.80	62.81	49.03	49.82	87.77	55.33	
OCR (ECCV20) [14] †	26.63	75.95	66.81	68.65	63.23	62.12	50.60	50.68	87.96	55.60	
PCNet (CVPR20) [62]	33.94	76.65	67.07	68.74	60.22	60.16	47.65	48.67	88.68	57.03	
CNIF (ICCV19) [43]	30.20	76.38	68.28	69.49	65.52	65.51	52.67	53.38	87.99	57.74	
NPPNet(ICCV21) [22]	33.54	75.31	71.04	71.50	71.75	70.55	56.66	55.84	88.92	58.56	
SCHP (PAMI20) [58]	32.83	76.63	69.33	71.76	67.93	67.42	56.56	57.55	88.40	59.36	
PAGRnet-HRNetV2-W48 (ours) †	29.81	75.79	70.06	71.84	70.22	68.93	57.24	57.68	88.06	58.39	
PAGRnet-HRNetV2-W48 (ours)	32.19	76.87	71.65	73.14	71.47	70.22	58.45	58.73	88.67	59.85	

[†] means single-scale testing results without flipping. | **bold** – the highest value.

except [43], [58], [62], [63] whose results are based on a large input size. Due to hardware restriction, we only calculated our results based on input size 384×384 , unlike most published work for other methods using 473×473 .

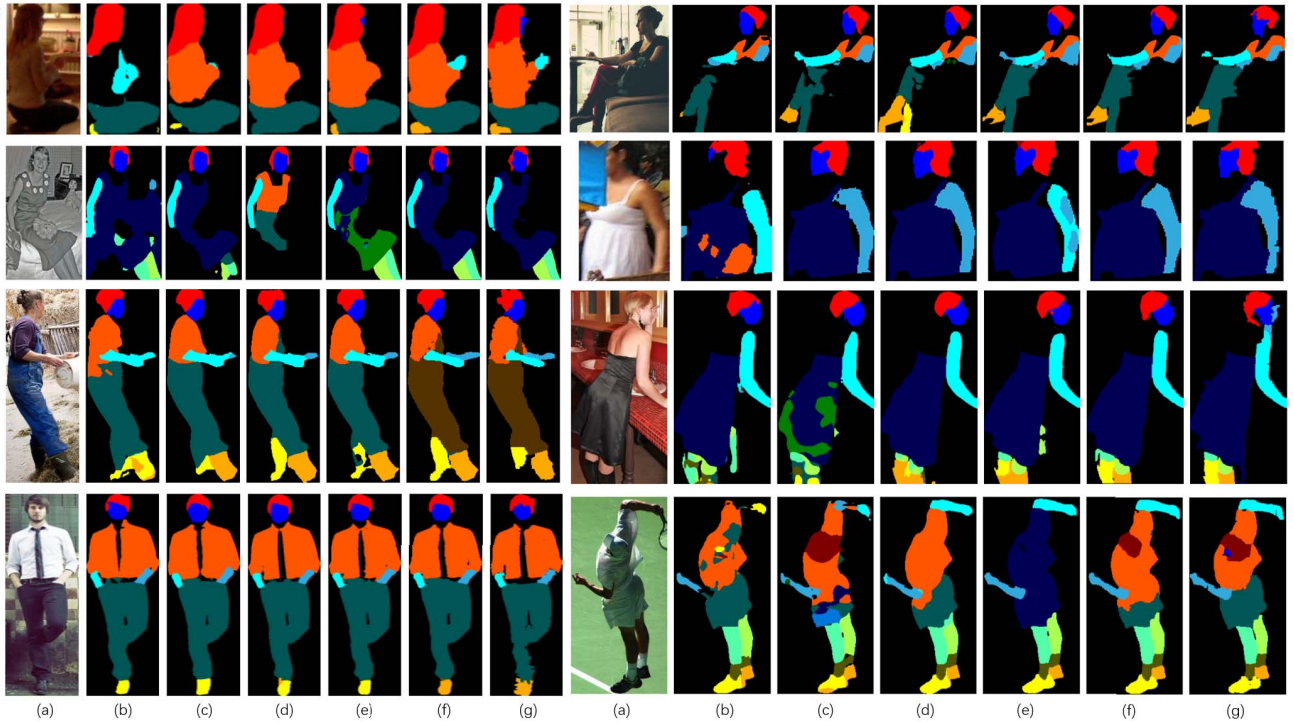


Fig. 7. Qualitative comparison of human parsing results: (a) input images, (b) CE2P [34] results, (c) CorrPM [19] results, (d) OCR results [67], (e) SCHP [58], (f) our results, and (g) ground truths.

Following similar studies [1], [43], [58], [63], we applied multi-scale (MS) and left-right flipping (Flip) to refine the results while testing. Table III shows that by applying MS of (0.75; 1.0; 1.25) and Flip improved the performance of our model based on HRNetV2-W48 from 58.39% to 59.85%, outperforming all other methods listed in Table III. First, our method outperforms OCR [14] a large margin by 3.85%; OCR [14] is also based on HRNetV2-W48 and it explores global representation from semantic segmentation prediction while we explore global representation from pose estimation. Moreover, with regard to methods using edge information, our scheme outperforms CE2P [34] by 6.75%, PGEcNet [39] by 5.51% and PCNet [62] by 2.82%, even though the input size of our model is 384×384 that is smaller than these schemes of 473×473 . Lastly, with our simple yet effective pose-based global representation learning module, our method achieves 2.11% improvement over CNIF [43], which uses complex hierarchy network structure and iterative training. Compared with NPPNet [22] in Table III, our method outperforms in terms of average mean IoU by 1.28%. NPPNet uses individual networks to learn human parsing and pose estimation, and add the interaction between their multi-scale features, so it can learn more discriminative features and obtain better performance on similar clothing items, like upper-clothing and coat. However, it is a search-based method, which requires many heuristics to work with different networks [73].

Compared to SCHP [58], our method outperforms in terms of average mean IoU by 0.49%. SCHP improves the reliability of parsing labels by aggregating the current learned model with the former sub-optimal ones iteratively, and it is effective to use the correlated labels to improve the performance of human

parsing. Unlike SCHP [58], which is more suitable for scenario with noisy labels, our method takes full advantage of the relationship between human parsing and pose estimation and exploits pose estimation to improve the performance of human parsing. In addition, our method takes less computing time and capacity during the training period than SCHP. Although with different focuses, our method has flexible structure that can also be trained using SCHP strategy for dataset with noisy data. We will demonstrate in later Section IV-E using Pascal-Person Part dataset, when pose ground-truths are not available, that our PAGRnet can be further improved by the integration with SCHP [58].

With reference to the per-class comparison shown in Table IV, the mean IoU for classes that are strongly related to human joints show substantial improvements in our method over the best per-class results among most state-of-art methods, e.g., 76.65% vs. 76.87% for the face, 71.04% vs. 71.65% for the left arm, 71.76% vs. 73.14% for the right arm, 56.66% vs. 58.73% for the left shoe, 57.55% vs. 58.73% for the right shoe.

2) *Qualitative Comparison:* We present some of our results from the LIP validation set in Fig. 7. Compared to the methods of CE2P [34], CorrPM [19], OCR [14], SCHP [58] models, our method yields more precise predictions. For example, in the last row on the left, our method correctly labels the right shoe, while all other methods identify it as left shoe. Our model also provides fine-grained details for human body parts. In the first row on the left, the region of the left arm is successfully segmented by our method. In the third row on the right, again our method can successfully segment the left and right legs as well as left and right shoes. In general,

TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE TESTING SET OF PASCAL-PERSON-PART DATASET

Methods	Pose	Head	Torso	U-Arm	L-Arm	U-Leg	L-Leg	Bkg	Ave.
Attention (CVPR16) [68]		81.47	59.06	44.15	42.50	38.28	35.62	93.65	56.39
SS-JPPNet (PAMI18) [1]	✓	83.26	62.40	47.80	45.58	42.23	39.48	94.68	59.36
Attention+MMAN (ECCV18) [36] [†]		82.58	62.83	48.49	47.37	42.80	40.40	94.92	59.91
Joint (arXiv15) [18]	✓	85.50	67.87	54.72	54.30	48.25	44.76	95.32	64.69
DeepLabV2 (PAMI2017) [28]		-	-	-	-	-	-	-	64.94
MuLA (ECCV18) [45]	✓	84.60	68.30	57.50	54.10	49.60	46.40	95.60	65.10
DeepLabV3+ (ECCV18) [69]		87.02	72.02	60.37	57.36	53.54	48.52	96.07	67.84
WSHP (CVPR18) [70]	✓	87.15	72.28	57.07	56.21	52.43	50.36	97.72	67.60
SPGNet (ICCV19) [71]		87.67	71.41	61.69	60.35	52.62	48.80	95.98	68.36
PGN (ECCV18) [72]		90.89	75.12	55.83	64.61	55.42	41.57	95.33	68.40
PGEcNet (TCSVT21) [39]		87.35	71.70	61.70	61.62	52.56	51.65	95.85	68.92
SMDC (TC21) [46] [†]	✓	88.25	74.34	58.94	64.54	55.39	51.83	95.89	69.88
CNIF (ICCV19) [43]		88.02	72.91	64.31	63.52	55.61	54.96	96.02	70.76
SCHP (PAMI20) [58]		87.41	73.80	64.98	64.70	57.43	55.62	96.26	71.46
NPPNet (ICCV21) [22]	✓	86.78	72.50	66.33	63.95	58.06	58.59	95.88	71.73
CDCL (TCSVT21) [20]	✓	75.53	66.26	63.28	57.14	47.75	51.45	93.72	65.02
CDCL+Pascal (TCSVT21) [20]	✓	86.39	74.70	68.32	65.98	59.86	58.70	95.79	72.82
PAGRnet-ResNet101 (ours)	✓	86.84	74.32	63.50	59.61	54.19	51.51	96.01	69.43
PAGRnet-HRNetV2-W48 (ours)	✓	87.68	76.14	65.70	62.94	57.05	54.09	96.27	71.41
PAGRnet+SCHP-HRNetV2-W48 (ours)	✓	87.41	76.05	66.41	63.46	57.22	55.83	96.14	71.79

[†] means single-scale testing results without flipping.

by effectively exploiting pose-aware global representation, our approach outputs reasonable results for labels that are closely related to human joints in a human parsing task.

E. Evaluation on the Pascal-Person-Part Dataset

Apart from the LIP dataset, we also compare our method using the Pascal-Person-Part [23] and the ATR [35] datasets. Table V shows our results on the Pascal-Person-Part dataset [23] in comparison with 17 major state-of-the-art methods. Table V shows that our model based on ResNet101 achieves a mean IoU of 69.43%, which outperforms most of the methods that integrate pose information for human part segmentation except CDCL+Pascal [20] and SMDC [46]. For instance, based on ResNet101, our method outperforms SS-JPPNet [1] by 10.13%, Joint [18] by 4.47%, MuLA [45] by 4.33%, WSHP [70] by 1.83% and PGEcNet [39] by 0.51%. Although our method only slightly outperforms NPPNet [22], NPPNet exploits accurate pose annotations and extra edge information. It is important to note that NPPNet is a search-based method and requires many heuristics to work well [73]. The mean IoU of CNIF [43] is higher than that of our method, because the Pascal-Person-Part dataset involve with data of extreme human poses and the quality of estimated poses in our implementation (obtained using pose estimator [55]) was not very satisfactory.

By using the more powerful backbone HRNetV2-W48, our method obtains 71.41% which outperforms CNIF [43]. Following SCHP [58], we apply the model aggregation and label refinement that uses the soft predictions as the ground-truths to train the model after 100 epochs, after which our method obtains 71.79% mean IoU. This has demonstrated that reducing the pose noises and improving the estimated pose quality can further improve the parsing performance, which will be studied in our future work.

The reason why CDCL+Pascal [20] performs better than our method is that it exploits additional large number of

synthetic human images and COCO-DensePose [74] dataset for training. Compared to the CDCL model only trained with synthetic and COCO-DensePose images, our method outperforms by 4.41% based on ResNet101 backbone and by 6.93% based on HRNetV2-W48. Since synthetic images can obtain the segmentation and pose annotations easily, and our PAGRnet can also learn the human pose information that can effectively bridge the synthetic and real domains during training [20], we can also use synthetic images to further improve the performance. However, our focus in this paper is to exploit the global pose information for human parsing. The discussion on the use of extra dataset is beyond the scope of this paper and this will be covered in the future work by another paper. Moreover, CDCL [20] can only address the problem of person part segmentation, in which one-to-one correspondence between segmentation labels and pose labels is required. Therefore, CDCL [20] method is not applicable to other datasets, like LIP and ATR. Compared to SMDC [46], our method slightly underperforms by 0.45%. However, SMDC [46] uses both pose and edge information, and larger input size (512×512) for training. In addition, SMDC [46] requires to pretrain the pose model in advance, and the pose prediction and parsing prediction can not be trained simultaneously in one network.

F. Evaluation on the ATR Dataset

Table VI shows the evaluation results on the randomly partitioned testing set of ATR dataset [35] (see Section IV-A). We compare our proposed PAGRnet with 11 major state-of-the-art methods in the table, except [75] because the testing set of [75] was a selected one. Table VI shows that our model outperforms others on most of the evaluation metrics. Compared to TGPNet [57], PAGRnet based on ResNet101 obtains 85.63% F1-score, which brings 3.87% improvement. Our PAGRnet based on ResNet101 also outperforms CNIF [43] and SCHP [58] on all metrics. Compared to PGEcNet [39],

TABLE VI
COMPARISON OF HUMAN PARSING PERFORMANCE METRICS (%) ON THE ATR TESTING SET

Models	Backbone	Pixel acc.	Foreground acc.	Precision	Recall	F1-score
ATR (PAMI15) [56]	Alexnet	91.11	71.04	71.69	60.25	64.38
DeepLab2 (PAMI17) [9]	ResNet101	94.42	82.93	78.48	69.24	73.53
PSPNet (CVPR17) [10]	ResNet101	95.20	80.23	79.66	73.79	75.84
Attention (CVPR16) [68]	ResNet101	95.41	85.71	81.30	73.55	77.23
DeepLab3+ (ECCV18) [69]	ResNet101	95.96	83.04	80.41	78.79	79.49
Co-CNN (ICCV15) [35]	Alexnet	96.02	83.57	84.95	77.66	80.14
TGPNNet (ACMM18) [57]	ResNet101	96.45	87.91	83.36	80.22	81.76
CorrPM (CVPR20) [19] $\triangle$	ResNet101	95.80	86.45	81.28	84.39	82.80
PGECNet (TCSVT2020) [39]	ResNet101	97.03	89.01	86.61	84.31	85.23
CNIF (ICCV19) [43]	ResNet101	96.26	87.91	84.62	86.41	85.51
SCHP (PAMI20) [58]	ResNet101	96.25	87.97	84.99	86.13	85.55
PAGRnet-ResNet101 (ours)	ResNet101	96.39	88.81	85.07	86.19	85.63
PAGRnet-HRNetV2-W48 (ours)	HRNetV2-W48	96.41	88.77	86.00	86.44	86.21

$\triangle$ the result is by running the shared model of CorrPM [19] on our data split, not their reported result.

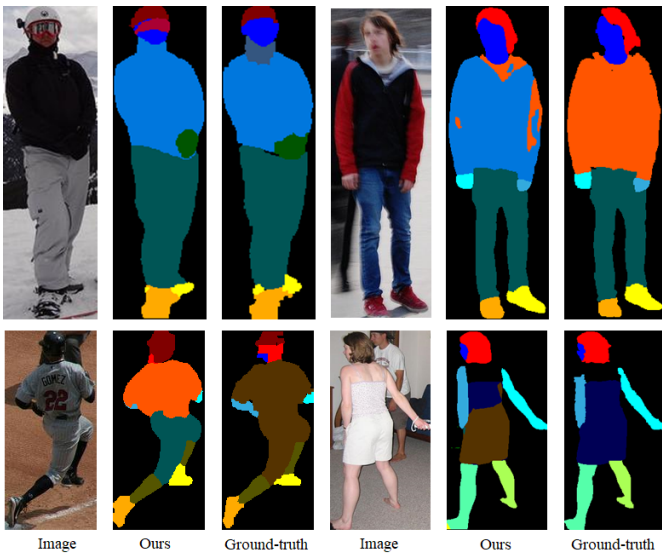


Fig. 8. Selected failure cases of the LIP validation set.

our method outperforms in terms of F1-score and recall, but underperforms in terms of pixel accuracy, foreground accuracy and precision. On the basis of the more powerful backbone HRNetV2-W48, PAGRnet achieves 86.21% F1-score, which outperforms CNIF [43] and SCHP [58] by 0.70% and 0.66%, respectively.

It is worthy to note that our results are obtained by estimating human pose by pose estimator “Detectron2” [55]; this demonstrates another advantage of our method: it is flexible for application and can be deployed without precise pose information, saving the cost of manual annotation.

G. Error Analysis and Limitation

Fig. 8 shows two types of failure cases for rare labels and similar labels. As shown, our proposed model will have slight performance degradation on the rare classes, like scarf, glasses or jumpsuit. It can be seen from the first column of Fig. 8 that the scarf is not identified in the first row example and the jumpsuit is confused in the second row example. Besides, our method may produce inferior results for some clothing

items with similar appearance, such as dress versus skirt and upper clothing versus coat. As shown in the second column of Fig. 8, the upper-clothing is confused with coat in the first row example and the dress is confused with jumpsuit in the second row example. The reason is that, in this paper, we focus on exploiting pose estimation to improve parsing and use a very simple encoder, which may be not enough to learn a powerful representation to identify rare labels or similar labels. We will work on address these issues in the future work.

V. CONCLUSION AND FUTURE WORK

In this paper, we explored to learn global information from pose estimation and apply to improve a human parsing task. We proposed a pose-aware global representation to augment the parsing feature. Our pose-aware global representation learning scheme uses a global joint representation and the parsing pixel-joint relationship to generate a pose-aware representation. We evaluated our approach on three benchmark datasets, and showed that it outperformed other methods that integrate pose estimation for parsing, such as CorrPM [19], in terms of accuracy and efficiency. Our approach achieved competitive performance on benchmark datasets, namely 59.85% on the LIP [1] validation set, 71.79% on the Pascal-Person-Part [23] testing set, and 86.21% the ATR [35] dataset. The effectiveness of our method has been demonstrated through extensive quantitative and qualitative comparisons.

The key contribution of this paper is that we propose a novel way of aggregating global information; more specifically, we exploit *global* information from pose estimation to assist the learning of human parsing. This helps shed light on other multi-task learning problems and can be viewed a possible solution for other tasks or applications, where global information is crucial, e.g., person identification [76].

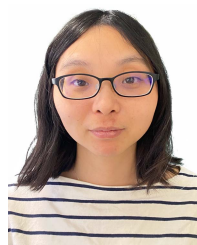
In future work, we will explore to improve the human parsing and pose estimation simultaneously, even on noisy datasets.

REFERENCES

- [1] X. Liang, K. Gong, X. Shen, and L. Lin, “Look into person: Joint body parsing & pose estimation network and a new benchmark,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 4, pp. 871–885, Apr. 2018.

- [2] S. Verma, S. Anand, C. Arora, and A. Rai, "Diversity in fashion recommendation using semantic parsing," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 500–504.
- [3] K. Yamaguchi, M. H. Kiapour, L. E. Ortiz, and T. L. Berg, "Parsing clothing in fashion photographs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 3570–3577.
- [4] J. Guo, Y. Yuan, L. Huang, C. Zhang, J.-G. Yao, and K. Han, "Beyond human parts: Dual part-aligned representations for person re-identification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3642–3651.
- [5] M. Tian et al., "Eliminating background-bias for robust person re-identification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5794–5803.
- [6] B. Su, J. Zhou, X. Ding, H. Wang, and Y. Wu, "Hierarchical dynamic parsing and encoding for action recognition," in *Proc. Eur. Conf. Comput. Vis. Cham, Switzerland: Springer*, 2016, pp. 202–217.
- [7] Y. Wang, D. Tran, Z. Liao, and D. Forsyth, "Discriminative hierarchical part-based models for human parsing and action recognition," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 3075–3102, 2012.
- [8] D. Guan, Y. Cao, J. Yang, Y. Cao, and M. Y. Yang, "Fusion of multispectral data through illumination-aware deep neural networks for pedestrian detection," *Inf. Fusion*, vol. 50, pp. 148–157, Oct. 2019.
- [9] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.
- [10] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
- [11] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 3146–3154.
- [12] X. Liu, L. Zhu, S. Zhu, and L. Luo, "Distance guided channel weighting for semantic segmentation," 2020, *arXiv:2004.12679*.
- [13] F. Zhang et al., "ACFNet: Attentional class feature network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2019, pp. 6798–6807.
- [14] Y. Yuan, X. Chen, X. Chen, and J. Wang, "Segmentation transformer: Object-contextual representations for semantic segmentation," 2019, *arXiv:1909.11065*.
- [15] J. Fan, B. Liu, K. Zhang, and Q. Liu, "Semi-supervised video object segmentation via learning object-aware global-local correspondence," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Jul. 19, 2021, doi: [10.1109/TCSVT.2021.3098118](https://doi.org/10.1109/TCSVT.2021.3098118).
- [16] A. Toshev and C. Szegedy, "DeepPose: Human pose estimation via deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1653–1660.
- [17] J. Dong, Q. Chen, X. Shen, J. Yang, and S. Yan, "Towards unified human parsing and pose estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 843–850.
- [18] F. Xia, J. Zhu, P. Wang, and A. Yuille, "Pose-guided human parsing with deep learned features," 2015, *arXiv:1508.03881*.
- [19] Z. Zhang, C. Su, L. Zheng, and X. Xie, "Correlating edge, pose with parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8900–8909.
- [20] K. Lin, L. Wang, K. Luo, Y. Chen, Z. Liu, and M.-T. Sun, "Cross-domain complementary learning using pose for multi-person part segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 1066–1078, Mar. 2020.
- [21] Y. Yang, Y. Zhuang, and Y. Pan, "Multiple knowledge representation for big data artificial intelligence: Framework, applications, and case studies," *Frontiers Inf. Technol. Electron. Eng.*, vol. 22, no. 12, pp. 1551–1558, 2021.
- [22] D. Zeng, Y. Huang, Q. Bao, J. Zhang, C. Su, and W. Liu, "Neural architecture search for joint human parsing and pose estimation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 11385–11394.
- [23] X. Liang, X. Shen, J. Feng, L. Lin, and S. Yan, "Semantic object parsing with graph LSTM," in *Proc. Eur. Conf. Comput. Vis. Cham, Switzerland: Springer*, 2016, pp. 125–143.
- [24] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
- [25] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Jan. 2017.
- [26] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1925–1934.
- [27] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.
- [28] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
- [29] X. Sun, C. Chen, X. Wang, J. Dong, H. Zhou, and S. Chen, "Gaussian dynamic convolution for efficient single-image segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 5, pp. 2937–2948, May 2021.
- [30] X. Weng, Y. Yan, S. Chen, J.-H. Xue, and H. Wang, "Stage-aware feature alignment network for real-time semantic segmentation of street scenes," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 7, pp. 4444–4459, Jul. 2021.
- [31] H. Zhao et al., "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 267–283.
- [32] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
- [33] H. Sang, Q. Zhou, and Y. Zhao, "PCANet: Pyramid convolutional attention network for semantic segmentation," *Image Vis. Comput.*, vol. 103, Nov. 2020, Art. no. 103997.
- [34] T. Ruan, T. Liu, Z. Huang, Y. Wei, S. Wei, and Y. Zhao, "Devil in the details: Towards accurate single and multiple human parsing," in *Proc. AAAI Conf. Artif. Intell.*, vol. 33, Jul. 2019, pp. 4814–4821.
- [35] X. Liang et al., "Human parsing with contextualized convolutional neural network," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 1386–1394.
- [36] Y. Luo, Z. Zheng, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Macro-micro adversarial network for human parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
- [37] W. Ji, X. Li, F. Wu, Z. Pan, and Y. Zhuang, "Human-centric clothing segmentation via deformable semantic locality-preserving network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 12, pp. 4837–4848, Dec. 2019.
- [38] E. Huang, Z. Su, F. Zhou, and R. Wang, "Learning rebalanced human parsing model from imbalanced datasets," *Image Vis. Comput.*, vol. 99, Jul. 2020, Art. no. 103928.
- [39] S. Zhang, G.-J. Qi, X. Cao, Z. Song, and J. Zhou, "Human parsing with pyramidal gather-excite context," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 1016–1030, Mar. 2020.
- [40] Y.-T. Chan, S.-J. Wang, and C.-H. Tsai, "Real-time foreground detection approach based on adaptive ensemble learning with arbitrary algorithms for changing environments," *Inf. Fusion*, vol. 39, pp. 154–167, Jan. 2018.
- [41] B. Zhu, Y. Chen, M. Tang, and J. Wang, "Progressive cognitive human parsing," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 7607–7614.
- [42] Y. Zhou, P. Y. Mok, and S. Zhou, "A part-based deep neural network cascade model for human parsing," *IEEE Access*, vol. 7, pp. 160101–160111, 2019.
- [43] W. Wang, Z. Zhang, S. Qi, J. Shen, Y. Pang, and L. Shao, "Learning compositional neural information fusion for human parsing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5703–5713.
- [44] T. Li, Z. Liang, S. Zhao, J. Gong, and J. Shen, "Self-learning with rectification strategy for human parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9263–9272.
- [45] X. Nie, J. Feng, and S. Yan, "Mutual learning to adapt for joint human parsing and pose estimation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 502–517.
- [46] B. Yang, C. Yu, J.-G. Yu, C. Gao, and N. Sang, "Pose-guided hierarchical semantic decomposition and composition for human parsing," *IEEE Trans. Cybern.*, early access, Sep. 21, 2021, doi: [10.1109/TCYB.2021.3107544](https://doi.org/10.1109/TCYB.2021.3107544).
- [47] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 325–341.
- [48] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "Learning a discriminative feature network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1857–1866.
- [49] Y. Yuan, L. Huang, J. Guo, C. Zhang, X. Chen, and J. Wang, "OCNet: Object context network for scene parsing," 2018, *arXiv:1809.00916*.

- [50] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [51] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.
- [52] J. Tompson, R. Goroshin, A. Jain, Y. LeCun, and C. Bregler, "Efficient object localization using convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 648–656.
- [53] B. Xiao, H. Wu, and Y. Wei, "Simple baselines for human pose estimation and tracking," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 466–481.
- [54] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5693–5703.
- [55] Y. Wu, A. Kirillov, F. Massa, W.-Y. Lo, and R. Girshick. (2019). *Detectron2*. [Online]. Available: <https://github.com/facebookresearch/detectron2>
- [56] X. Liang et al., "Deep human parsing with active template regression," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 12, pp. 2402–2414, Mar. 2015.
- [57] X. Luo, Z. Su, J. Guo, G. Zhang, and X. He, "Trusted guidance pyramid network for human parsing," in *Proc. 26th ACM Int. Conf. Multimedia*, Oct. 2018, pp. 654–662.
- [58] P. Li, Y. Xu, Y. Wei, and Y. Yang, "Self-correction for human parsing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 6, pp. 3260–3271, Jun. 2020.
- [59] H. Zhang et al., "Context encoding for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160.
- [60] X. Liang, L. Lin, X. Shen, J. Feng, S. Yan, and E. P. Xing, "Interpretable structure-evolving LSTM," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1010–1019.
- [61] S. Xie, R. Girshick, P. Dollar, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1492–1500.
- [62] X. Zhang, Y. Chen, B. Zhu, J. Wang, and M. Tang, "Part-aware context network for human parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8971–8980.
- [63] W. Wang, H. Zhu, J. Dai, Y. Pang, J. Shen, and L. Shao, "Hierarchical human parsing with typed part-relation reasoning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8929–8939.
- [64] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Apr. 2019.
- [65] J. Zhao et al., "Self-supervised neural aggregation networks for human parsing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2017, pp. 7–15.
- [66] X. Liu, M. Zhang, W. Liu, J. Song, and T. Mei, "BraidNet: Braiding semantics and details for accurate human parsing," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 338–346.
- [67] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. 16th Eur. Conf. Comput. Vis. Glasgow, U.K.: Springer*, Aug. 2020, pp. 173–190.
- [68] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3640–3649.
- [69] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
- [70] H.-S. Fang, G. Lu, X. Fang, J. Xie, Y.-W. Tai, and C. Lu, "Weakly and semi supervised human body part parsing via pose-guided knowledge transfer," 2018, *arXiv:1805.04310*.
- [71] B. Cheng et al., "SPGNet: Semantic prediction guidance for scene parsing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 5218–5228.
- [72] K. Gong, X. Liang, Y. Li, Y. Chen, M. Yang, and L. Lin, "Instance-level human parsing via part grouping network," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 770–785.
- [73] B. Zoph and Q. V. Le, "Neural architecture search with reinforcement learning," 2016, *arXiv:1611.01578*.
- [74] R. A. Güler, N. Neverova, and I. Kokkinos, "DensePose: Dense human pose estimation in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7297–7306.
- [75] K. Gong, Y. Gao, X. Liang, X. Shen, M. Wang, and L. Lin, "Graphonomy: Universal human parsing via graph transfer learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7450–7459.
- [76] J. Liu, B. Ni, Y. Yao, P. Zhou, S. Cheng, and J. Hu, "Pose transferrable person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4099–4108.



Her current research interests include deep learning, image understanding, and image segmentation.



fashion applications, advanced data analysis, and artificial intelligent applications in the fashion industry.

Yanghong Zhou received the bachelor's degree from Fujian Normal University in 2011, the master's degree from the University of Electronic Science and Technology of China in 2014, and the Ph.D. degree focusing on the research of fashion image understanding from The Hong Kong Polytechnic University, Hong Kong, in 2019. After graduation, she is currently a Post-Doctoral Fellow at the Laboratory for Artificial Intelligence in Design, Hong Kong Science Park. This work was largely based on the work of Dr. Yanghong Zhou in her Ph.D. study.

P. Y. Mok (Member, IEEE) received the B.Eng. (Hons.) and Ph.D. degrees in industrial and manufacturing systems engineering from The University of Hong Kong in 1998 and 2002, respectively. She is currently an Associate Professor with The Hong Kong Polytechnic University. Her current research interests include fashion pattern engineering, fashion 2D and 3D CAD, digital human modeling, 3D scanning and sizing, cloth simulation, deep learning, computer generated textile, sketch and pattern designs, computer vision and computer graphics in fashion applications, advanced data analysis, and artificial intelligent applications in the fashion industry.